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GCSE · OCR · Computer Science

3 hours 50 questions

Designing, Creating & Refining Algorithms

Designing, Creating & Refining Algorithms

Inputs, Processes, & Outputs for a Problem / Structure Diagrams / Pseudocode & Flowcharts / Identify Errors in Algorithms / Trace Tables

Easy (12 questions)	/24
Medium (17 questions)	/70
Hard (21 questions)	/109
Total Marks	**/203**

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Easy Questions

1 **Write pseudocode** to increment the value held in the variable score by one.

..... **(1 mark)**

2 A fast food restaurant offers half-price meals if the customer is a student or has a discount card. The offer is not valid on Saturdays. A computer system is used to identify whether the customer can have a half-price meal.

The table identifies the three inputs to the computer system:

Input	Value
A	Is a student
B	Has a discount card
C	The current day is Saturday

The restaurant adds a service charge to the cost of a meal depending on the number of people at a table. If there are more than five people 5% is added to the total cost of each meal.

Customers can also choose to leave a tip, this is optional and the customer can choose between a percentage of the cost, or a set amount.

Identify all the **additional inputs** that **will be required** for this change to the algorithm.

.....

..... **(2 marks)**

3 (a) **Write a pseudocode statement** to assign the value 7.3 to a variable with the **identifier** timer

..... **(1 mark)**

(b) **State** the most **appropriate data type** for the variable timer

..... **(1 mark)**

4 (a) The algorithm for one section of a vending machine program is shown in pseudocode.

```
if money >= price then
    venditem()
    giveChange(money – price)
else
    print("Error – not enough money inserted")
endif
```

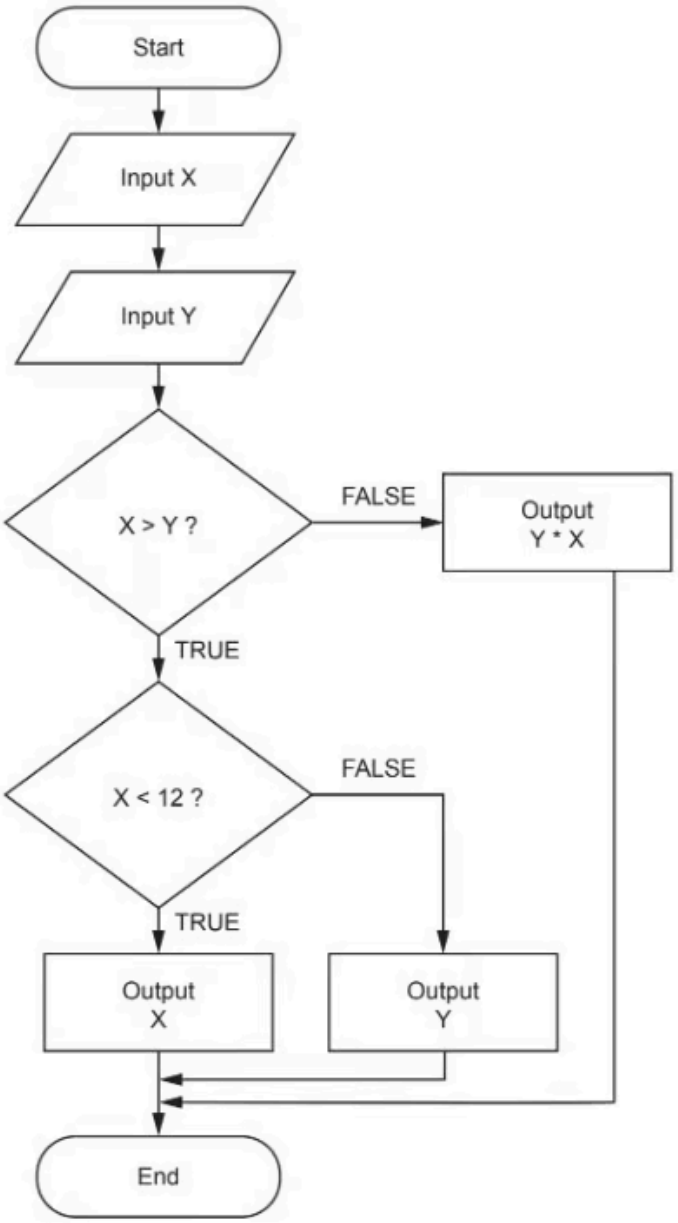
Give the identifier of **one variable** used in the algorithm.

..... **(1 mark)**

(b) **State how many parameters** are passed into the giveChange() subroutine.

..... **(1 mark)**

5 A programmer creates an algorithm using a flow chart.



Complete the table to give the output when each of the following set of values are input into the algorithm as X and Y.

Input value of X	Input value of Y	Output
15	10	
6	5	
2	3	
12	2	

(4 marks)

6 (a) A programmer has written an algorithm to output a series of numbers. The algorithm is shown below:

```
for k = 1 to 3
    for p = 1 to 5
        print (k + p)
    next p
next k
m = 7
print m * m
```

Describe what is meant by a **variable**.

.....

..... **(2 marks)**

(b) **Identify two variables** that have been used in the algorithm above.

.....

..... **(2 marks)**

7 A program creates usernames for a school. The first design of the program is shown in the flowchart in **Fig. 2**.

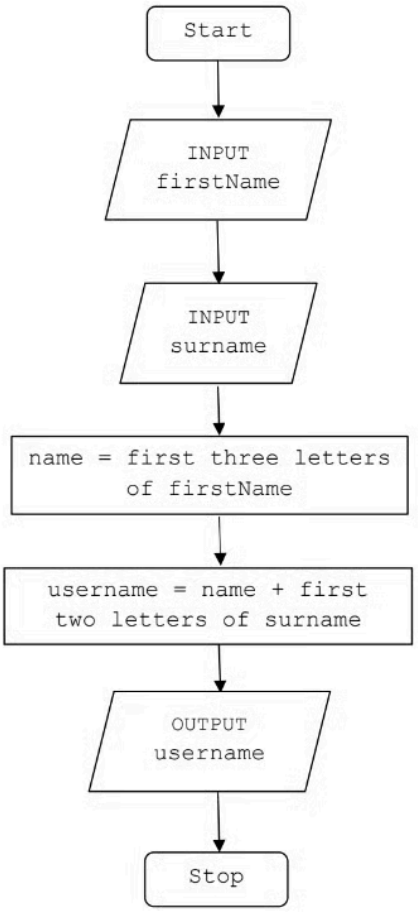


Fig. 2

For example, using the process in Fig. 2, Tom Ward's username would be TomWa.

State, using the process in **Fig. 2**, the username for **Rebecca Ellis**.

..... **(1 mark)**

8 Give **one** example of a syntax error in a program.

..... **(1 mark)**

9 A program allows users to search for and watch videos.

Users give a rating to the videos they watch.

Identify one input and one output for the program.

.....

..... **(2 marks)**

10 **Define** the term “syntax error”

.....

..... **(2 marks)**

11 **Define** the term 'logic error'

.....

..... **(2 marks)**

12 **Identify** one tool a programmer can use to manually find logic errors in an algorithm

..... **(1 mark)**

Medium Questions

1 A hotel car park charges £4 per hour. If the car is electric, this price is halved to £2 per hour.

Write an algorithm to:

- take as input the number of hours the user has parked and whether their car is electric or not
- calculate and output the total price
- repeat continually until the user enters 0 hours.

You must use **either**:

- OCR Exam Reference Language, or
- a high level programming language that you have studied.

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..... **(6 marks)**

2 (a) A program needs to perform the following tasks:

- Input two numbers from the user
- Compare both numbers and output the largest number.

Complete the pseudocode for this program.

```
num1 = ..................................................

num2 = input("enter second number")

........................... num1 > ........................... then

    ..................................................

else

    ..................................................

endif
```

(5 marks)

(b) A second program needs to perform the following tasks:

- Input a number from the user
- Double the number input and print the result
- Repeat bullets 1 and 2 until the user enters a number less than 0.

Write an algorithm for this program.

.....

.....

.....

.....

..... **(5 marks)**

3 The following program uses a **condition-controlled loop**.

```
x = 15

y = 0

while x > 0

    y = y + 1

    x = x – y

endwhile

print(y)
```

Complete the trace table to test this program.

x	y	output

(4 marks)

4 Ali's computer uses virtual memory. Ali has written two procedures to help himself understand how virtual memory works.

storeData() describes how data is stored in RAM.
accessData() describes how data is read from RAM

Write the letter of the missing statements from the table in the correct place to complete the algorithms.

Not all statements are used, and **some statements might be used more than once**.

```
procedure storeData()

  if RAM is ........................ then
```

```
        move data from RAM to ...............................
    endif
    store data in next free space in .........................
..................................................
procedure accessData()
    if ..................... (data required is in RAM) then
        if RAM is full then
            move unneeded data from RAM to HDD
        endif
        move required data from HD to RAM
    endif
    read data from ...................
endprocedure
```

Letter	Statement
A	Secondary storage
B	NOT
C	Full
D	endfunction
E	Empty
F	endprocedure
G	AND
H	RAM

(6 marks)

5 Dru writes the following program using a high-level language.

```
01 function newscore(a,b)
02     temp = a*b
```

```
03     temp = temp + 1
04     return temp
05 endfunction
06 score = 18
07 name = "Dru"
08 print (score)
09 print ("name")
10 print (newscore(score,2))
11 print (score)
```

The following table contains the program code for each line where this program outputs values.

State the values output by the program on each of the lines.

Line	Program code	Value output
08	print (score)	
09	print ("name")	
10	print (newscore(score,2))	
11	print (score)	

.....

.....

.....

..... **(4 marks)**

6 A vending machine can be in one of three states: on, off or suspended. A user can change the state of the vending machine by using the following algorithm.

```
newstate = input("Enter the new state : ")

switch newstate:

    case "on":

            statevalue = 1

    case "off":

            statevalue = 2

    case "suspended":

            statevalue = 3

    default: print("Invalid state")

endswitch
```

Rewrite the algorithm to perform the same actions **using IF statements** in place of the switch statement.

.....

.....

.....

.....

..... **(5 marks)**

7 (a) A programmer has written an algorithm to output a series of numbers. The algorithm is shown below:

```
01 for k = 1 to 3
02     for p = 1 to 5
03         print (k + p)
04     next p
05 next k
06 m = 7
07 print m * m
```

Give the first three numbers that will be printed by this algorithm

..... **(1 mark)**

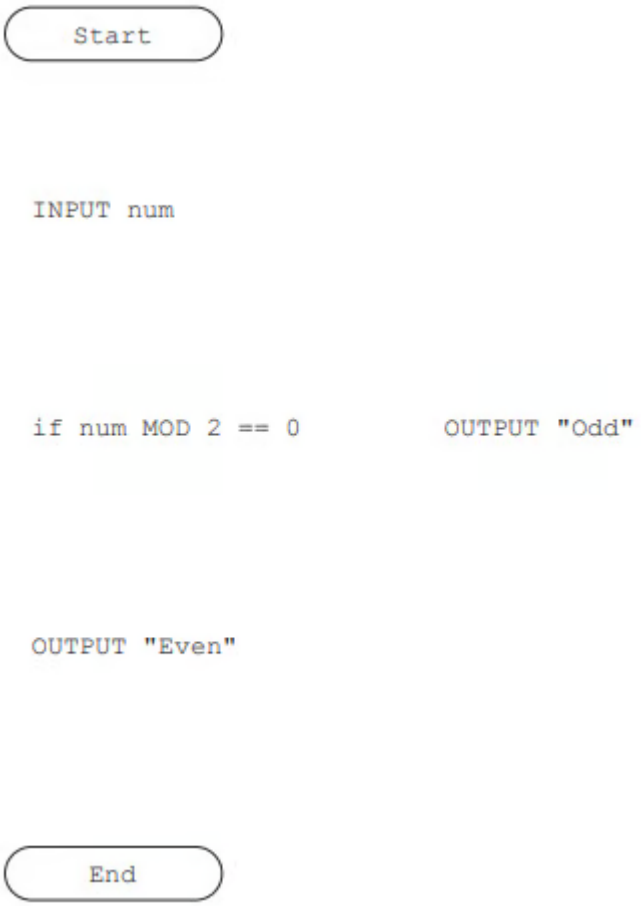
(b) **State how many times** line **03** will be executed if the algorithm runs through **once**.

..... **(1 mark)**

8 An algorithm decides if a number is odd or even. An odd number divided by 2 will give the remainder 1.

The flowchart statements have been written for the algorithm, but the flowchart is incomplete.

Complete the flowchart.



(4 marks)

9 A traffic light system can be in one of four states: red, amber, green or flashing. A traffic controller can change the state of the traffic light by using the following algorithm.

```
newlight = input("Enter the new light state : ")

switch newlight:

    case "red":

        lightcode = 1

    case "amber":

        lightcode = 2

    case "green":

        lightcode = 3

    case "flashing":

        lightcode = 4

    default: print("Invalid light state")

endswitch
```

Rewrite the algorithm to perform the same actions **using IF statements** in place of the switch statement.

.....

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..... **(5 marks)**

10 A student writes:

```
num1 = input(Enter a number)
```

Explain why this line contains a syntax error

.....

..... **(2 marks)**

11
```
01 user = input(Enter name)
02 if user == "Admin" then
03     print("Welcome")
04 endif
```

Identify the line number of the syntax error and **explain** why it will prevent the program from running

.....

.....

..... **(3 marks)**

12
```
if age > 11 or age < 18 then
  print("Secondary Student")
```

Describe the logic error in this statement

.....

..... **(2 marks)**

13 **Explain** why logic errors are often more difficult to find than syntax errors

.....

..... **(2 marks)**

14
```
01 num = input("Enter number")
02 if num > 10 OR num < 20 then
03   print("Number is valid")
```

Describe the logic error in line 02 and **identify** the output if the user enters 25

.....

.....

..... **(3 marks)**

15 The following program uses a **condition-controlled loop**.

```
a = 20

b = 1

while a >= b

    a = a - b

    b = b + 2

endwhile

print(b)
```

Complete the trace table to test this program.

a	b	output

(4 marks)

16 The following program uses a **condition-controlled loop**.

```
x = 10

y = 2

while x > 0

    x = x - y

endwhile

print(x)
```

Complete the trace table to test this program.

x	y	output

(4 marks)

17 A local library uses a computer system to allow members to manage their accounts.

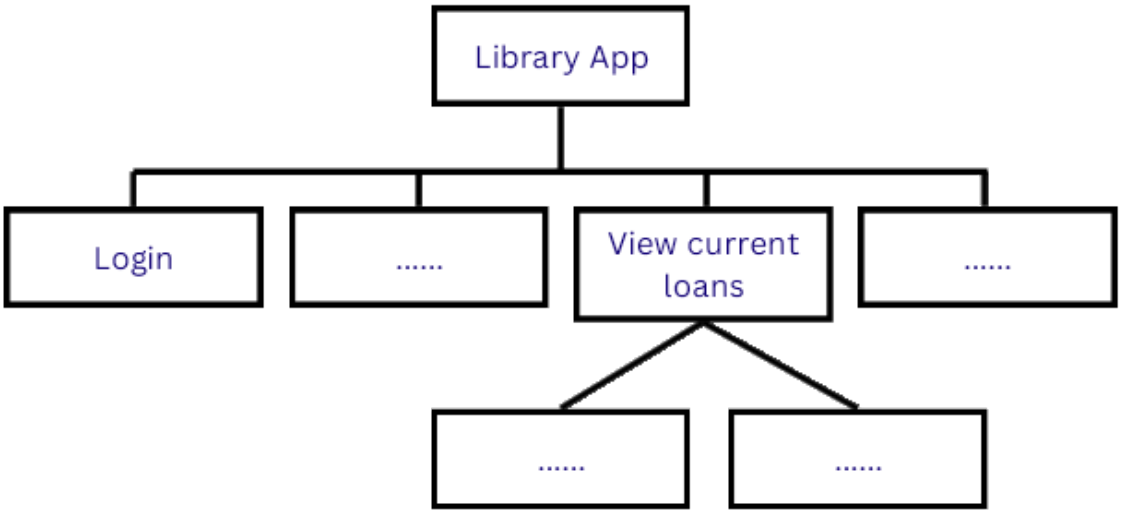
Members must **log in** before accessing the main menu. From the main menu, they can **search for books**, **view current loans**, or **pay outstanding fines**. If they choose to view current loans, they can then choose to **renew a book** or **return a book**.

Task:

A structure diagram has been used to design the library system.

Write one letter from the following table in each space to complete the diagram.

Letter	Task
A	Renew a book
B	Pay outstanding fines
C	Search for books
D	Return a book



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..... **(4 marks)**

Hard Questions

1 A fast food restaurant offers half-price meals if the customer is a student or has a discount card.

The offer is not valid on Saturdays.

A computer system is used to identify whether the customer can have a half-price meal.

The table identifies the three inputs to the computer system:

Input	Value
A	Is a student
B	Has a discount card
C	The current day is Saturday

The restaurant needs an algorithm designing to help employees work out if a customer can have a half price meal or not. It should:

- input required data
- decide if the customer is entitled to a discount
- output the result of the calculation.

Design the algorithm using a flowchart.

(5 marks)

2 Each member of staff that works in a restaurant is given a Staff ID. This is calculated using the following algorithm.

```
01    surname = input("Enter surname")

02    year = input("Enter starting year")

03    staffID = surname + str(year)

04    while staffID.length < 10

05        staffID = staffID + "x"

06    endwhile

07    print("ID " + staffID)
```

Complete the following trace table for the given algorithm when the surname "**Kofi**" and the year **2021** are entered.

You may **not need** to use all rows in the table.

Line number	surname	year	staffID	Output
01	Kofi			
02		2021		

(4 marks)

3 Jack is writing a program to add up some numbers. His first attempt at the program is shown.

```
a = input("Enter a number")

b = input("Enter a number")

c = input("Enter a number")

d = input("Enter a number")

e = input("Enter a number")

f = (a + b + c + d + e)

print(f)
```

Jack decides to improve his program. He wants to be able to input how many numbers to add together each time the algorithm runs, and also wants it to calculate and display the average of these numbers.

Write an algorithm to:

- ask the user to input the quantity of numbers they want to enter and read this value as input
- repeatedly take a number as input, until the quantity of numbers the user input has been entered
- calculate and output the total of these numbers
- calculate and output the average of these numbers.

(6 marks)

4 When a new booking is recorded, the details are entered into a program to validate the values. The following criteria are checked:

- firstName and surname are not empty
- room is either “basic” or “premium”
- nights is between 1 and 5 (inclusive).

If any invalid data is found “NOT ALLOWED” is displayed. If all data is valid “ALLOWED” is displayed.

Complete the following program to validate the inputs.

```
firstName = input("Enter a first name")

surname = input("Enter a surname")

room = input("Enter basic or premium")

nights = input("Enter between 1 and 5 nights")

stayComplete = False
```

You must use either:

- OCR Exam Reference Language, or
- a high-level programming language that you have studied.

(5 marks)

5 (a) A program creates usernames for a school. The first design of the program is shown in the flowchart in **Fig. 2**.

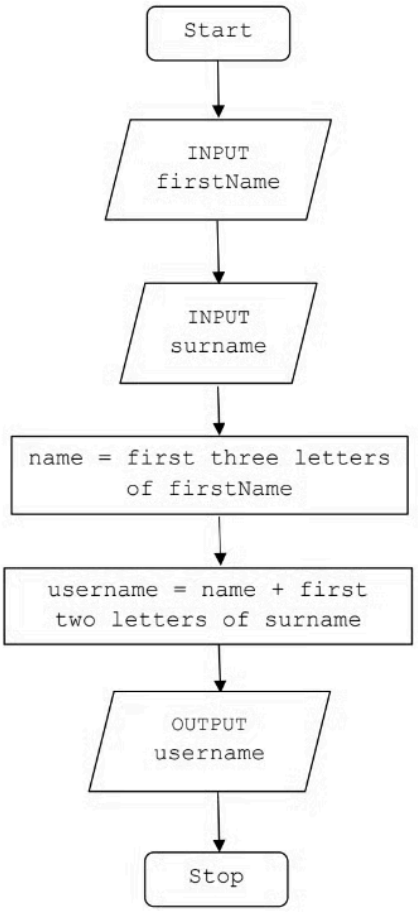

Fig. 2

The program design is **updated** to create usernames as follows:

- If the person is a teacher, their username is the last 3 letters of their surname and then the first 2 letters of their first name.
- If the person is a student, their username is the first 3 letters of their first name and then the first 2 letters of their surname.

What would be the username for a teacher called **Fred Biscuit** using the **updated process**?

..... **(1 mark)**

(b) **Write an algorithm** for the updated program design shown in part(a).

.....

.....

.....

.....

.....

..... **(6 marks)**

6 (a) Data for one week (Monday to Friday) is stored in a 2D array with the identifier minsPlayed.

The following table shows part of this array, containing 4 students.

			Students			
			Stuart	Wes	Victoria	Dan
			0	1	2	3
Days of the week	Mon	0	60	30	45	0
	Tue	1	180	60	0	60
	Wed	2	200	30	0	20
	Thu	3	60	10	15	15
	Fri	4	100	35	30	45

The teacher writes a program to add up and print out the total number of minutes student 2 played computer games over 5 days (Monday to Friday).

```
total = 0

total = total + minsPlayed[2,0]

total = total + minsPlayed[2,1]

total = total + minsPlayed[2,2]

total = total + minsPlayed[2,3]

total = total + minsPlayed[2,4]

print(total)
```

Refine the program to be **more efficient**. Write the refined version of the algorithm.

You must use **either**:

- OCR Exam Reference Language, **or**
- a high-level programming language that you have studied.

.....

.....

..... **(4 marks)**

(b) The following flowchart outputs a message depending on how long each person has spent playing computer games.

Rewrite the flowchart as a program.

You must use **either**:

- OCR Exam Reference Language, **or**
- a high-level programming language that you have studied.

.....

.....

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..... **(4 marks)**

7 Taylor is writing an algorithm to record the results of an experiment.

Taylor needs to be able to enter a numeric value which is added to a total which initially starts at 0.

Every time she enters a value, the total is output.

The algorithm repeats until the total is over 100.

Write an algorithm to implement Taylor's requirements.

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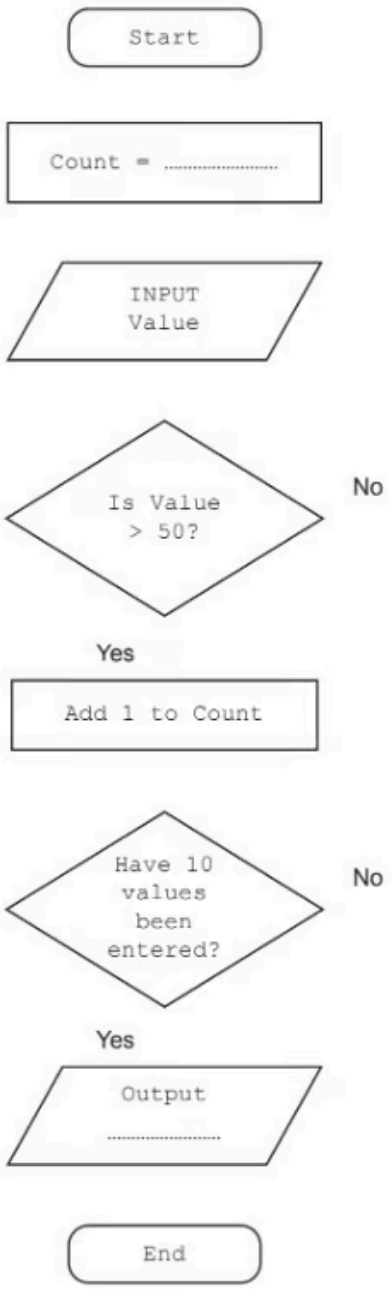
..... **(6 marks)**

8 (a) For the next part of the experiment, Taylor needs to be able to enter 10 values and count how many of the values are over 50, outputting this value once all values have been entered.

Complete the following flowchart to implement this algorithm



(5 marks)

(b) **Write a pseudocode algorithm** that uses iteration to allow Taylor to:

- enter 10 values
- count how many values are over 50
- output the count of values over 50 after all 10 values are entered.

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..... **(5 marks)**

9 OCRBlocks is a game played on a 5 × 5 grid. Players take it in turns to place blocks on the board. The board is stored as a two-dimensional (2D) array with the identifier gamegrid

Fig. 6.1 shows that players A and B have placed three blocks each so far

	0	1	2	3	4
0	A			B	
1					
2		B			
3	A		B		
4			A		

Fig. 6.1

Write an algorithm to allow player A to select a position for their next block on the game board.

The algorithm must:

- ask the player for the position of their block on the board
- use the checkblock() function to check if this position is free
- if the position is free, add the letter "A" to the position chosen in the gamegrid array
- if the position is not free, repeat the above steps until a free position is chosen.

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..... **(6 marks)**

10 The algorithm for one section a the vending machine program is shown in pseudocode

```
if money >= price then
   venditem()
   giveChange(money – price)
else
   print("Error – not enough money inserted")
endif
```

Draw the vending machine algorithm as a **flowchart**.

(5 marks)

11 The following names of students are stored in an array with the identifier studentnames.

studentnames = ["Rob", "Anna", "Huw", "Emma", "Patrice", "Iqbal"]

A school uses the array to call an attendance register every morning.

Write an algorithm using **iteration** to:

- display the name of each student one at a time from studentnames
- take as input whether that student is present or absent
- display the total number of present students and number of absent students in a suitable message, after all student names have been displayed.

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..... **(6 marks)**

12 (a) One encryption method is a Caesar cipher.

This Caesar cipher moves each letter of the alphabet **one** place to the right.

The following table shows the original letters in the first row, and the new letters in the second row.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A

For example, if the message read: HELLO

This would be stored as: IFMMP

The following pseudocode algorithm takes a string of uppercase letters as input and uses the Caesar cipher to encrypt them.

The functions used in the algorithm are described in the table:

Function	Description
ASC(character)	Returns the ASCII value for character e.g. ASC("A") returns 65
CHR(ASCIIvalue)	Returns the single character for ASCIIvalue e.g. CHR(65) returns "A"
subString(Value, Number)	Returns the Number of characters starting at position Value (where 0 is the first character)

Complete the pseudocode algorithm to perform a Caesar cipher.

```
01 message = input("Please enter your string")

02 newMessage = ""

03 messageLength = message.length

04 for count = 0 to..........................

05     ASCIIValue = ASC(message.subString(..............,1))
```

```
06    ASCIIValue = ASCIIValue + .........

07    if ASCIIValue >90 then

08        ASCIIValue = ............................... – 26

09    endif

10    newMessage = ..................... + CHR(ASCIIValue)

11 next count
```

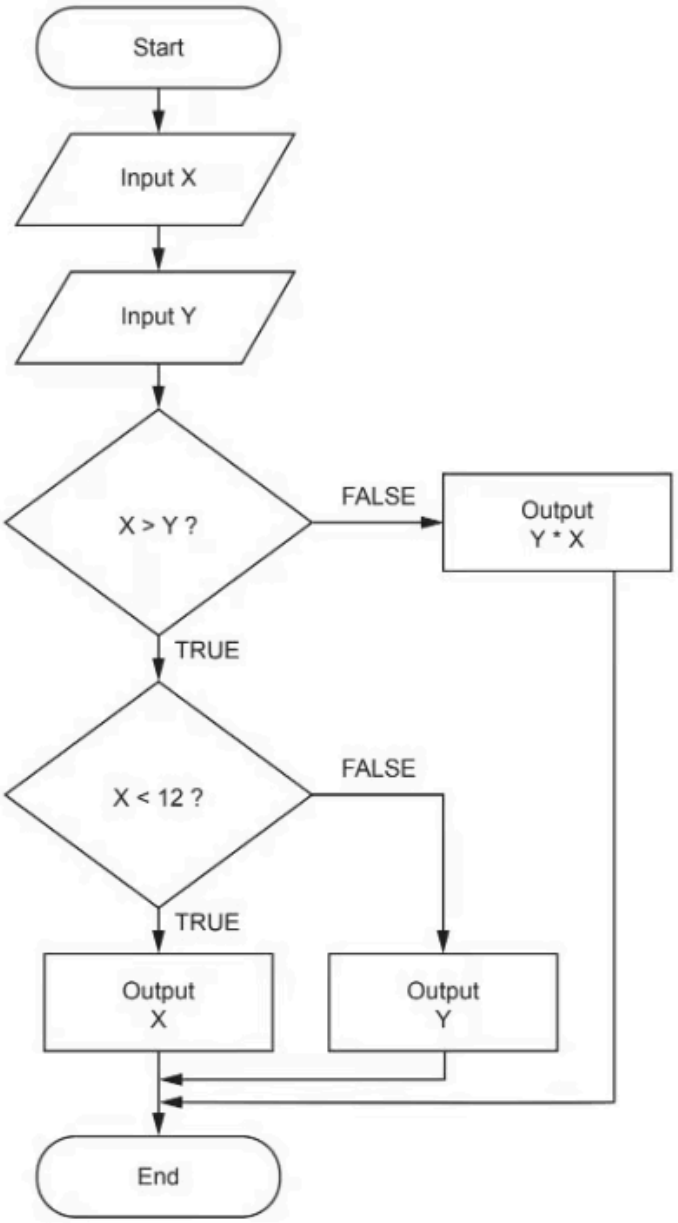
(5 marks)

(b) The algorithm needs adapting. An extra line (line 12) is needed to output the encrypted message.

Write line 12 to **output the encrypted message** in pseudocode or programming code.

..... **(1 mark)**

13 **Write** this algorithm using **pseudocode**.



(6 marks)

14 A single record from this database table is read into a program that uses an array with the identifier studentdata. An example of this array is shown below:

studentdata = [“Kirstie”, “Homework forgotten”, “-2”, “FALSE”]

The array is zero based, so studentdata[0] holds the value “Kirstie”.

Write an algorithm that will identify whether the data in the studentdata array shows that a letter has been sent home or not for the student. The algorithm should then output either “sent” (if a letter has been sent) or “not sent” (if a letter has not been sent).

(4 marks)

15 Using pseudocode, **write an algorithm** that will use a count-controlled loop to print out the numbers 1 to 10 in ascending order.

(3 marks)

16 OCR town are holding an election with three candidates (A, B and C). An electronic voting booth will be used to allow people to vote.

Write an algorithm that:

- Allows voters to enter either A, B or C.
- Keeps track of how many times each candidate has been voted for.
- As soon as one person has finished voting, allows the next person to vote.
- At any point allows the official to type in "END", which will print out the number of votes for each candidate and the total number of votes overall.

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..... **(6 marks)**

17 A student writes an algorithm to input two numbers and add them together to create a total.

If the total is between 10 and 20 inclusive, "success" is output.
If the total is not between 10 and 20 inclusive, "warning" is output.

```
01 num1 = input("Enter a number")
02 num2 = input("Enter a number")
03 total = num1 + num1
04 if total >= 10 then
05     print("success")
06 else
07     print("warning")
08 endif
```

The algorithm **does not work** correctly.

Identify the line number of the two logic errors in the algorithm and **refine** the code to correct each logic error.

.....

.....

.....

..... **(4 marks)**

18 This algorithm calculates the number of points a student gets for the distance they throw in the javelin:

```
01 javelinThrow = input("Enter distance")
02 yearGroup = input("Enter year group")
03 if javelinThrow >= 20.0 then
04     score = 3
05 elseif javelinThrow >= 10.0 then
06     score = 2
07 else
08     score = 1
09 endif
10 if yearGroup != 11 then
11     score = score * 2
12 endif
13 print("The score is", score)
```

Complete the trace table for the algorithm when a student in year 10 throws a distance of 14.3

You **may not** need to use all the rows in the table.

Line number	javelinThrow	yearGroup	score	Output

(4 marks)

19 **Explain** the impact of a syntax error on the compilation process

.....

.....

..... **(3 marks)**

20
```
01 for i = 0 to 5
02     print("Loop number " i)
03 next i
```
Identify the syntax error and **analyse** how it would be corrected in a high-level language

.....

.....

..... **(3 marks)**

21 total = total + total

Evaluate the impact of this line if it was intended to calculate a running total of items in a list

.....

..... **(2 marks)**